

COMP 3270 – Introduction to Algorithms

Homework 6

Sorting algorithms

1. [10 points] For the MERGE function, the sizes of the L and R arrays are one element longer than n_1 and n_2 , respectively. Can you rewrite the merge function with the size of L and R exactly equal to n_1 and n_2 ?
2. [5 points] In the MERGE function, auxiliary arrays L and R are used to store temporary values. How can you modify the function to perform merging in-place without using extra arrays?
3. [5 points] For QUICKSORT, the PARTITION function is called n times to sort an array of n elements. Prove that when the array is already sorted in ascending or descending order, every call to the PARTITION (A, p, r) function generates an empty array and an array of size $p - r$.
4. [5 points] Prove that if the input array is sorted in descending order and QuickSort always picks the last element as the pivot, it results in the worst-case runtime of $O(n^2)$.
5. [10 points] See below for Triple-QUICKSORT(A, p, r), which partitions an array into three subarrays A_1, A_2 , and A_3 , such that all elements in A_1 are less than those in A_2 , and all elements in A_2 are less than those in A_3 .

```
Triple-QUICKSORT( $A, p, r$ )
{
    if ( $p < r$ )
    {
        [ $q_1, q_2$ ] = Triple-PARTITION( $A, p, r$ );
        Triple-QUICKSORT ( $A, p, q_1-1$ );
        Triple-QUICKSORT ( $A, q_1+1, q_2-1$ );
        Triple-QUICKSORT ( $A, q_2+1, r$ );
    }
}
```

- a. Triple-PARTITION (A, p, r) returns two pivots q_1 and q_2 . When this function returns, all elements in $A[p \dots q_1-1]$ are less than or equal to the pivot $A[q_1]$, all elements in $A[q_1+1 \dots q_2-1]$ are larger than $A[q_1]$ and less than or equal to the pivot $A[q_2]$, and all elements in $A[q_2+1 \dots r]$ are larger than the pivot $A[q_2]$. Complete the code of Triple-PARTITION (A, p, r).

```
Triple-PARTITION( $A, p, r$ )
{  $x = A[r]$ 
```

```

i = p - 1
for j = p to r - 1
    if A[j] ≤ x
        i = i + 1
        exchange A[i] ↔ A[j]
exchange A[i + 1] ↔ A[r]
q1 = _____
x = A[r]
i = q1
for j = _____ to _____
    if A[j] ≤ _____
        i = i + 1

    exchange _____
    exchange _____
q2 = _____
return [q1, q2]
}

```

- b. Is it possible for Line 1 or line 9 of Triple-PARTITION choose the same element as the pivots (In other words, $q_1 = q_2$)? Please explain your reason.
 - c. Analyze the worst-case time complexity of Triple-QUICKSORT (A, p, r).
6. [5 points] Discuss the stability property in sorting algorithms and its significance. Explore how sorting algorithms may or may not maintain stability during the sorting process.
7. [10 points] Consider a scenario where you need to sort an array of integers with values in the range [0, 100]. Suggest and implement a linear time sorting algorithm suitable for this specific case. Analyze the time and space complexity of your algorithm.